

synth-attest: An Open, Vendor-Neutral Verifiable-Identity and Tamper-Evident Audit Layer for DNA-Synthesis Order Compliance

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Abstract—DNA-synthesis screening systems answer whether a sequence is dangerous, but not who ordered it, whether they are a verified researcher, or whether a trustworthy record of the decision exists. As regulation (the proposed US Biosecurity Modernization and Innovation Act, the US OSTP framework, UK DSIT guidance, and the proposed EU Biotech Act) converges on mandating customer identity verification and recordkeeping for synthesis providers, no shared, vendor-neutral tooling exists. Existing commercial entrants solve the problem with centralized, per-provider Know-Your-Customer workflows. We present *synth-attest*, an open-source layer that sits on top of (and never replaces) sequence screening, adding three capabilities built on W3C standards: a portable researcher-identity credential, a provider-issued revocable exemption credential modeled on existing exemption-certification practice, and a privacy-preserving, tamper-evident order-audit record. The design is vendor-neutral by construction: the core has no proprietary dependency and runs on a dependency-free reference engine, with any W3C Verifiable Credential backend pluggable behind a stable interface. We describe the threat model honestly, scoping the contribution to fraud, attribution, and deterrence at the acquisition node, and report a working prototype validated in continuous integration.

Index Terms—biosecurity, DNA synthesis, verifiable credentials, decentralized identifiers, audit trails, know-your-customer, compliance

I. INTRODUCTION

Advances in machine learning lower the barrier to designing biological sequences of concern, which makes the nucleic-acid synthesis-order step one of the few remaining physical chokepoints for governance. Sequence-screening systems such as SecureDNA [1] and the IBBIS Common Mechanism [2] assess whether an ordered sequence matches a hazard, but they do not establish *who* is ordering, whether that customer is a verified and legitimate researcher, or whether the order decision leaves a tamper-evident, independently verifiable record. Recent and proposed regulation, including the US Biosecurity Modernization and Innovation Act of 2026 [3], the US OSTP nucleic-acid-synthesis framework, UK DSIT screening guidance [4], and the proposed EU Biotech Act, pushes providers toward customer identity verification and recordkeeping. No shared, portable tooling exists to meet these obligations; commercial entrants address them with closed, per-provider Know-Your-Customer (KYC) workflows.

We present *synth-attest*, an open, vendor-neutral layer that adds the missing identity and audit capabilities on top of existing screening. It is deliberately scoped: it does not screen sequences and never handles sequence data.

II. THREAT MODEL AND HONEST SCOPE

We map the engineered-pandemic kill chain as: capability uplift, acquire materials, synthesize, weaponize, release. *synth-attest* intervenes at a single node, *acquire materials*, and only for actors routing through compliant providers. It addresses three failure modes at that node: identity fraud (spoofed or unverified customers), weak attribution (no portable record of who ordered what), and low deterrence. It explicitly does *not* address capability uplift, benchtop or non-compliant synthesis, weaponization, or a genuinely credentialed malicious insider; identity proves identity, not intent. Any claim that the layer “prevents engineered pandemics” is therefore unsupported and avoided. The system is one honest defense layer, complementary to sequence screening and benchtop controls.

III. DESIGN

synth-attest provides three artifacts and one privacy property.

A. Portable researcher-identity credential

A W3C Verifiable Credential [5] bound to a Decentralized Identifier [6] asserts a verified researcher and institutional affiliation only. It is explicitly not a clearance, and carries no risk-tiering field. Portability lets a researcher avoid re-onboarding at every provider, addressing the documented gap that customer screening is currently re-performed per provider and rarely shared.

B. Provider-issued exemption credential

Following the model of existing exemption-certification practice, an exemption credential is issued by the provider (the party that retains regulatory liability) to authorize a sequence *class*, never a sequence. It is time-bound and revocable. This separates the portable identity (about the person) from the

order-specific authorization (the provider’s local decision), correcting a common conflation between credentialing a person and clearing an order.

C. Privacy-preserving tamper-evident audit

Each order decision produces an append-only, signed record held by the provider. Only a salted hash commitment is anchored, and commitments are anchored in fixed-size padded batches, so that order volume, timing, and the screening verdict are not recoverable from what is anchored. This resolves the tension between tamper-evidence and metadata non-leakage: a verifier can confirm a record was not altered without learning order patterns. A customer-facing receipt never contains the screening verdict, preventing the audit channel from becoming a screening-evasion oracle.

D. Infohazard guard

No sequence-like data is permitted to enter the layer. A guard rejects any payload containing a nucleotide-alphabet run above a conservative threshold, erring toward false positives, which is the safe direction for an infohazard boundary.

IV. VENDOR NEUTRALITY

The core has no proprietary dependency. A dependency-free reference engine implements the full contract (issue, verify, detect tampering, revoke, selectively disclose), so the entire flow runs with only the Python standard library. A swappable `CredentialEngine` interface allows any conformant W3C VC/DID backend to be plugged in; an optional adapter for one such backend is provided but not required. Continuous integration enforces this: one job installs the core, asserts the optional backend is absent, and runs the suite; a second job validates the optional backend source-blind against its published package.

V. IMPLEMENTATION AND VALIDATION

The prototype implements all of the above plus reference adapters that map SecureDNA-shaped and IBBIS-Common-Mechanism-shaped screening results to opaque verdicts, and a custodial registry so that researchers need not operate a key wallet (a provider or neutral registry custodies the credential, keyed by a hashed identifier). Every credential and audit field is traced to a named regulatory obligation, each marked verified or unverified, and a test fails if any schema field lacks a mapping row. The test suite passes in continuous integration on Python 3.11 and 3.12, in both the no-proprietary-dependency configuration and the optional-backend configuration.

VI. RELATED WORK AND POSITIONING

Sequence-screening systems [1], [2] are complementary, not competing; synth-attest consumes their verdicts as opaque tokens. Commercial KYC entrants for synthesis providers use centralized identity verification and per-provider recordkeeping. The distinguishing properties of synth-attest are portable W3C credentials, cross-provider tamper-evident audit, and an open, vendor-neutral implementation, none of which the centralized approaches provide.

VII. LIMITATIONS AND FUTURE WORK

The exemption and audit models are validated against published descriptions of real screening practice but not yet against a live provider integration. The credential tiering question is deliberately omitted pending confirmation that any real framework tiers researcher clearance rather than data sensitivity. An issuer trust-root design (how identity issuers are accredited without recreating a centralized authority) is the principal open architectural question. Throughput at realistic order volumes and a live screening integration are the next validation steps.

VIII. CONCLUSION

synth-attest supplies the identity and audit layer that sequence screening leaves open, as an honest, narrowly scoped, and vendor-neutral open-source contribution to nucleic-acid synthesis governance.

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